

Project
of
Advance Automobile Technology
(MEN441)

BACHELOR OF TECHNOLOGY

In

MECHANICAL ENGINEERING



Session:- 2025-26

Submitted to:-
Dr. Rakesh Kumar Singh
Assistant Professor
Mechanical Engineering
DIT University ,Dehradun

Submitted by:
Harsh Pratap Singh
Roll NO:-23B13D0001
Sap ID:-1000019618
B.Tech. M.E.(3rd Year,5th
Sem

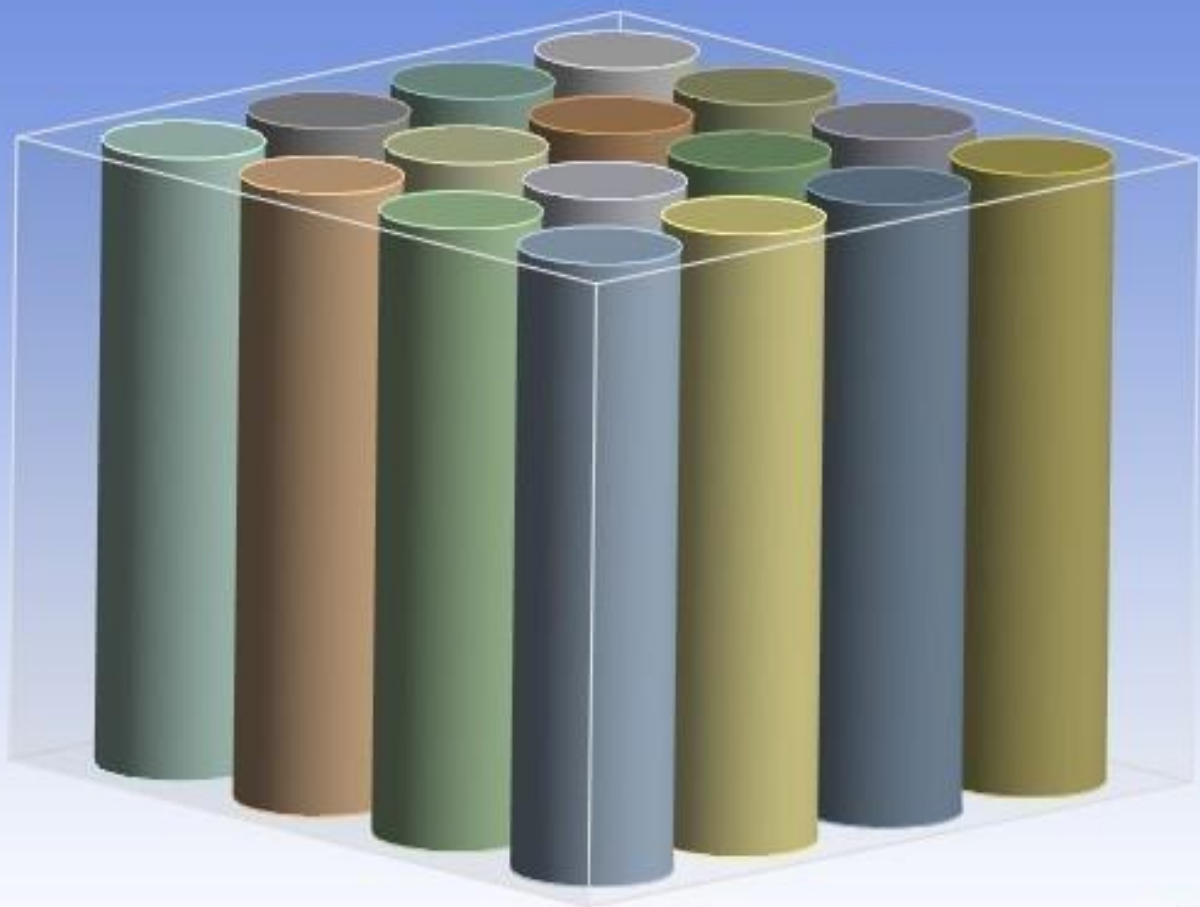


PROJECT TITLE

“Geometry Making of
Lithium - Ion Battery
Pack & CFD
Simulation in ANSYS”

Model

28/10/2025 12:28:08



0,00 50,00 (mm)
25,00



Contents

- [Units](#)
- [Model \(A3\)](#)
 - [Geometry Imports](#)
 - [Geometry Import \(A2\)](#)
 - [Geometry](#)
 - [Battery Pack](#)
 - [Cell](#)
 - [Parts](#)
 - [Materials](#)
 - [Coordinate Systems](#)
 - [Connections](#)
 - [Contacts](#)
 - [Contact Regions](#)
 - [Mesh](#)
 - [Named Selections](#)

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A3)

TABLE 2

Model (A3) > Geometry Imports

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3

Model (A3) > Geometry Imports > Geometry Import (A2)

Object Name	<i>Geometry Import (A2)</i>
State	Solved
Definition	
Source	D:\Battery Pack Simulation\16 Pack Battery_files\dp0\FFF\DM\FFF.agdb
Type	DesignModeler
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Advanced Geometry Options	
Compare Parts On Update	No
Analysis Type	3-D

Geometry

TABLE 4

Model (A3) > Geometry

Object Name	<i>Geometry</i>
State	Fully Defined
Definition	
Source	D:\Battery Pack Simulation\16 Pack Battery_files\dp0\FFF\DM\FFF.agdb
Type	DesignModeler

Length Unit	Meters
Bounding Box	
Length X	92, mm
Length Y	65, mm
Length Z	92, mm
Properties	
Volume	5,5016e+005 mm ³
Scale Factor Value	1,
Statistics	
Bodies	17
Active Bodies	17
Nodes	369900
Elements	333476
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	No

TABLE 5
Model (A3) > Geometry > Parts

Object Name	<i>Battery Pack</i>
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	0,1
Definition	
Suppressed	No
Coordinate System	Default Coordinate System
Treatment	None
Reference Frame	Lagrangian
Material	
Assignment	
Fluid/Solid	Defined By Geometry (Fluid)
Bounding Box	
Length X	92, mm
Length Y	65, mm
Length Z	92, mm
Properties	

Volume	2,8551e+005 mm³
Centroid X	46, mm
Centroid Y	32,5 mm
Centroid Z	46, mm
Statistics	
Nodes	112770
Elements	93324
Mesh Metric	None

FIGURE 1
Model (A3) > Geometry > Battery Pack > Figure

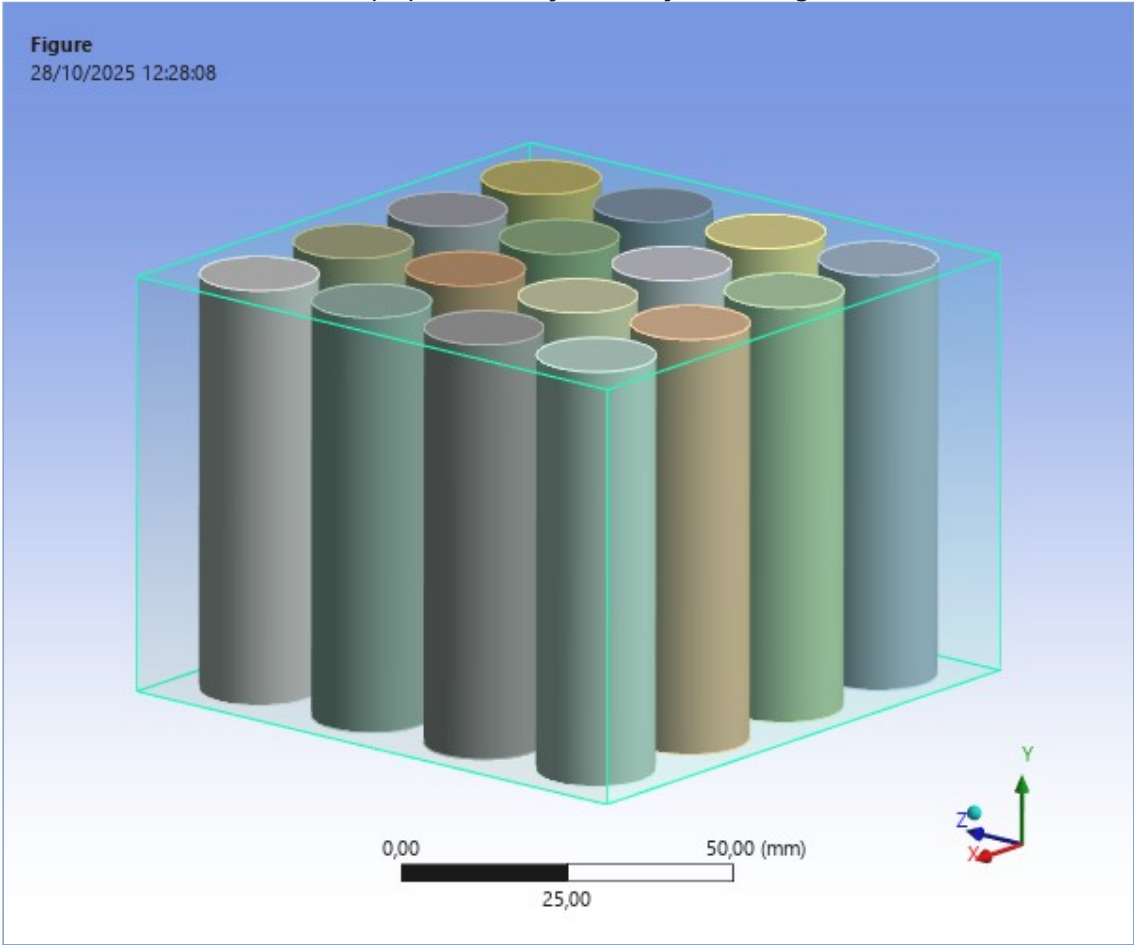


TABLE 6
Model (A3) > Geometry > Body Groups

Object Name	<i>Cell</i>
State	Meshed
Graphics Properties	
Visible	Yes
Definition	
Suppressed	No
Assignment	
Coordinate System	Default Coordinate System
Bounding Box	
Length X	84, mm
Length Y	65, mm
Length Z	84, mm
Properties	
Volume	2,6465e+005 mm³
Statistics	
Nodes	257130
Elements	240152

Mesh Metric

None

TABLE 7
Model (A3) > Geometry > Cell > Parts

Object Name	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6	Cell 7	Cell 8	Cell 9	Cell 10	Cell 11	
State	Meshed											
Graphics Properties												
Visible	Yes											
Transparency	1											
Definition												
Suppressed	No											
Coordinate System	Default Coordinate System											
Treatment	None											
Reference Frame	Lagrangian											
Material												
Assignment												
Fluid/Solid	Defined By Geometry (Solid)											
Bounding Box												
Length X	18, mm											
Length Y	65, mm											
Length Z	18, mm											
Properties												
Volume	16540 mm³											
Centroid X	13, mm	35, mm	57, mm	79, mm	13, mm	35, mm	57, mm	79, mm	13, mm	35, mm	57, mm	
Centroid Y	32,5 mm											
Centroid Z	13, mm				35, mm				57, mm			
Statistics												
Nodes	16110							15795		16110		
Elements	15048							14740		15048		
Mesh Metric	None											

TABLE 8
Model (A3) > Geometry > Cell > Parts

Object Name	Cell 12	Cell 13	Cell 14	Cell 15	Cell 16
State	Meshed				
Graphics Properties					
Visible	Yes				
Transparency	1				
Definition					
Suppressed	No				
Coordinate System	Default Coordinate System				
Treatment	None				
Reference Frame	Lagrangian				
Material					
Assignment					
Fluid/Solid	Defined By Geometry (Solid)				
Bounding Box					
Length X	18, mm				
Length Y	65, mm				
Length Z	18, mm				
Properties					
Volume	16540 mm ³				
Centroid X	79, mm	13, mm	35, mm	57, mm	79, mm
Centroid Y	32,5 mm				
Centroid Z	57, mm	79, mm			
Statistics					
Nodes	16110			15795	16110
Elements	15048			14740	15048

FIGURE 2
Model (A3) > Geometry > Cell > Figure

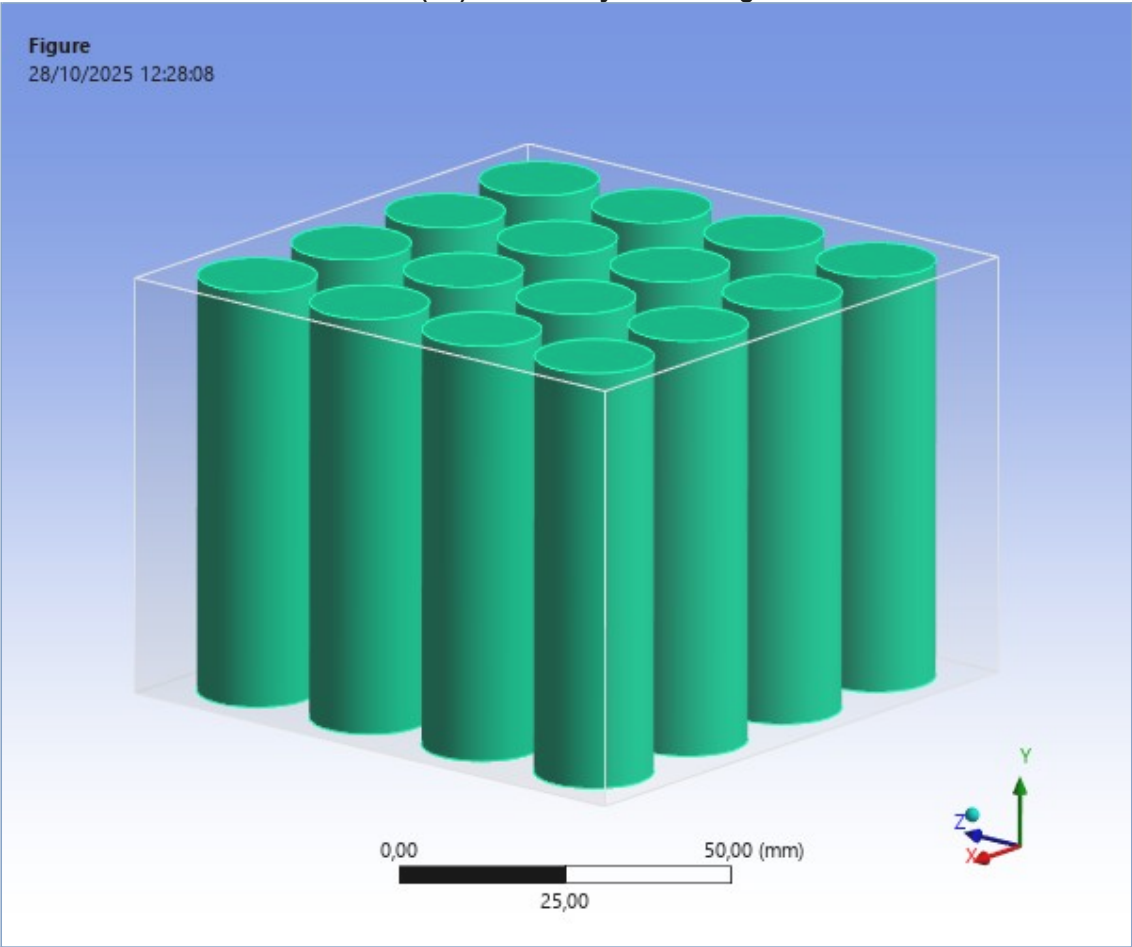


TABLE 9
Model (A3) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	0
Material Assignments	0

Coordinate Systems

TABLE 10
Model (A3) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0,
Origin	
Origin X	0, mm
Origin Y	0, mm
Origin Z	0, mm
Directional Vectors	
X Axis Data	[1, 0, 0,]
Y Axis Data	[0, 1, 0,]
Z Axis Data	[0, 0, 1,]
Transfer Properties	

Scope											
Scoping Method	Geometry Selection										
Contact	1 Face										
Target	1 Face										
Contact Bodies	Battery Pack										
Target Bodies	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6	Cell 7	Cell 8	Cell 9	Cell 10	Cell 11
Protected	No										
Definition											
Contact APDL Name											
Target APDL Name											
Display											
Element Normals	No										
Advanced											
Small Sliding	Program Controlled										

TABLE 14
Model (A3) > Connections > Contacts > Contact Regions

Object Name	Contact Region 12	Contact Region 13	Contact Region 14	Contact Region 15	Contact Region 16
State	Fully Defined				
Scope					
Scoping Method	Geometry Selection				
Contact	1 Face				
Target	1 Face				
Contact Bodies	Battery Pack				
Target Bodies	Cell 12	Cell 13	Cell 14	Cell 15	Cell 16
Protected	No				
Definition					
Contact APDL Name					
Target APDL Name					
Display					
Element Normals	No				
Advanced					
Small Sliding	Program Controlled				

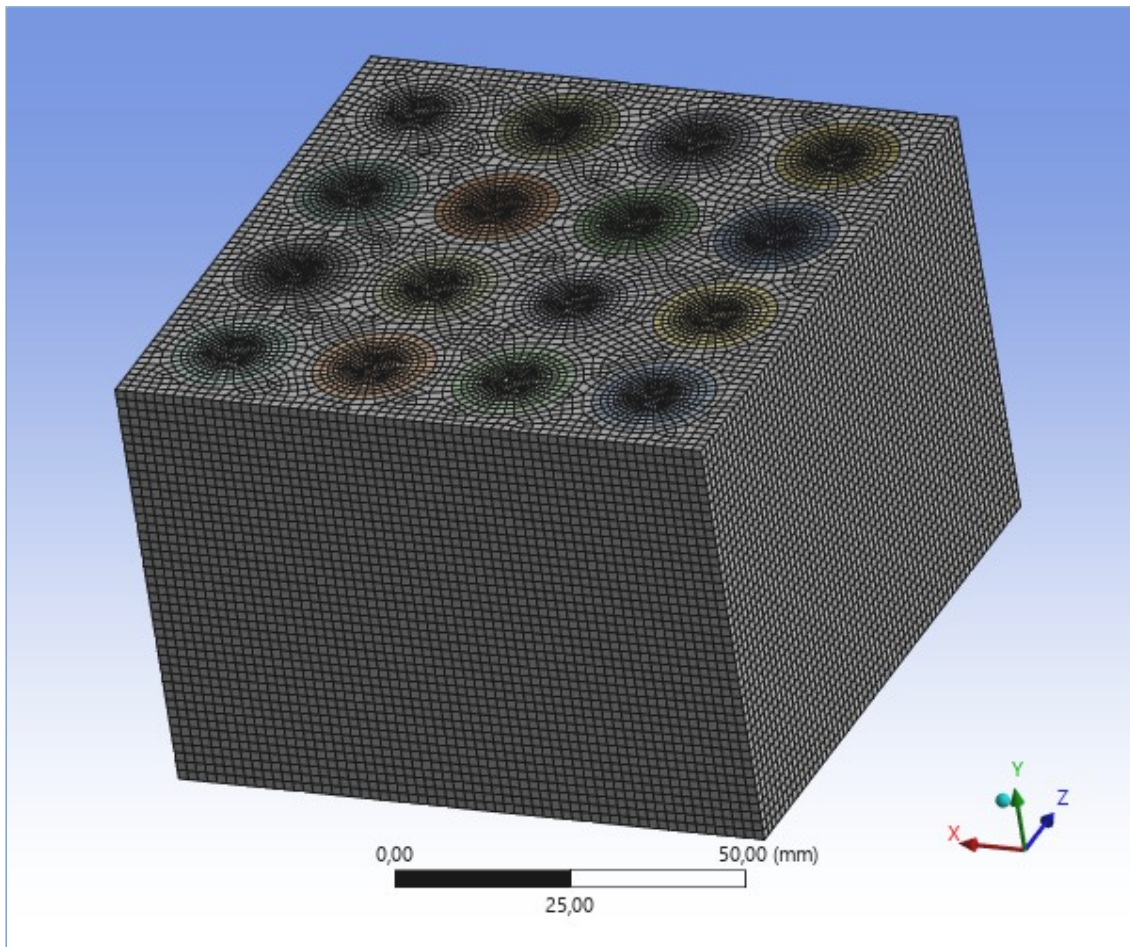
Mesh

TABLE 15
Model (A3) > Mesh

Object Name	Mesh
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	CFD
Solver Preference	Fluent
Element Size	1,5 mm
Export Format	Standard
Export Preview Surface Mesh	No
Sizing	

Use Adaptive Sizing	No
Growth Rate	Default (1,2)
Max Size	Default (3, mm)
Mesh Defeaturing	Yes
Defeature Size	Default (7,5e-003 mm)
Capture Curvature	Yes
Curvature Min Size	Default (1,5e-002 mm)
Curvature Normal Angle	Default (18,°)
Capture Proximity	No
Bounding Box Diagonal	145,44 mm
Average Surface Area	2261,7 mm ²
Minimum Edge Length	56,549 mm
Quality	
Check Mesh Quality	Yes, Errors
Target Skewness	Default (0,9)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0,272
Maximum Layers	5
Growth Rate	1,2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Default (1,35e-002 mm)
Generate Pinch on Refresh	No
Automatic Methods	
Sheet Body Method	Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	369900
Elements	333476
Show Detailed Statistics	No

FIGURE 3
Model (A3) > Mesh > Figure



Named Selections

TABLE 16
Model (A3) > Named Selections > Named Selections

Model (A0) Named Selections - Named Selections			
Object Name	Inlet	Outlet	Cell Boundary
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	1 Face	16 Faces	
Definition			
Send to Solver	Yes		
Protected	Program Controlled		
Visible	Yes		
Program Controlled Inflation	Exclude		
Statistics			
Type	Manual		
Total Selection	1 Face	16 Faces	
Surface Area	5980, mm ²	58736 mm ²	
Suppressed	0		
Used by Mesh Worksheet	No		

FIGURE 4
Model (A3) > Named Selections > Inlet > Figure 2

Figure 2

28/10/2025 12:28:09

 Inlet

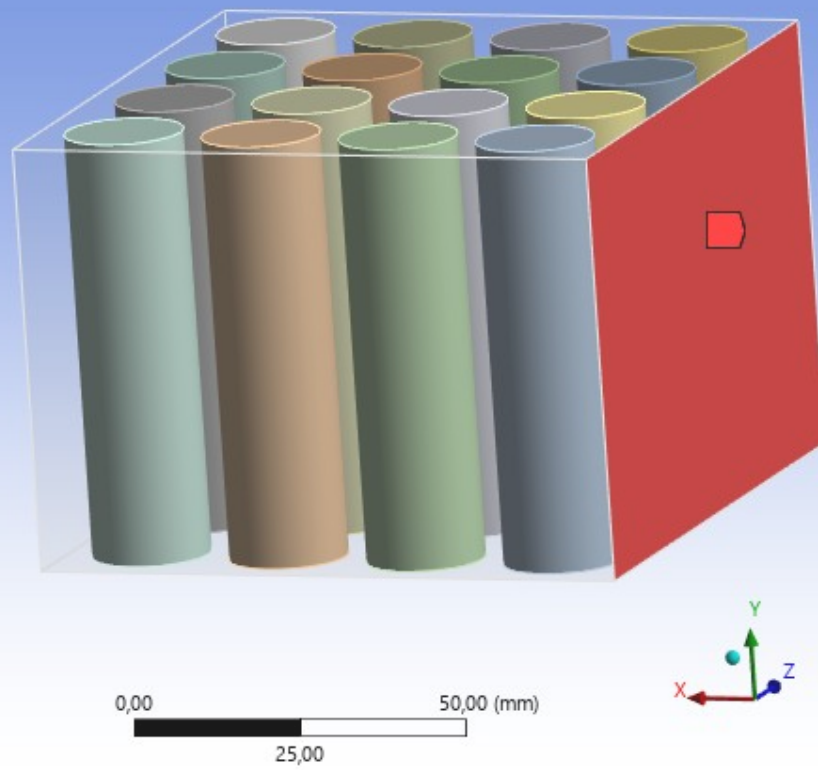


FIGURE 5

Model (A3) > Named Selections > Outlet > Figure 2

Figure 2

28/10/2025 12:28:09

Outlet

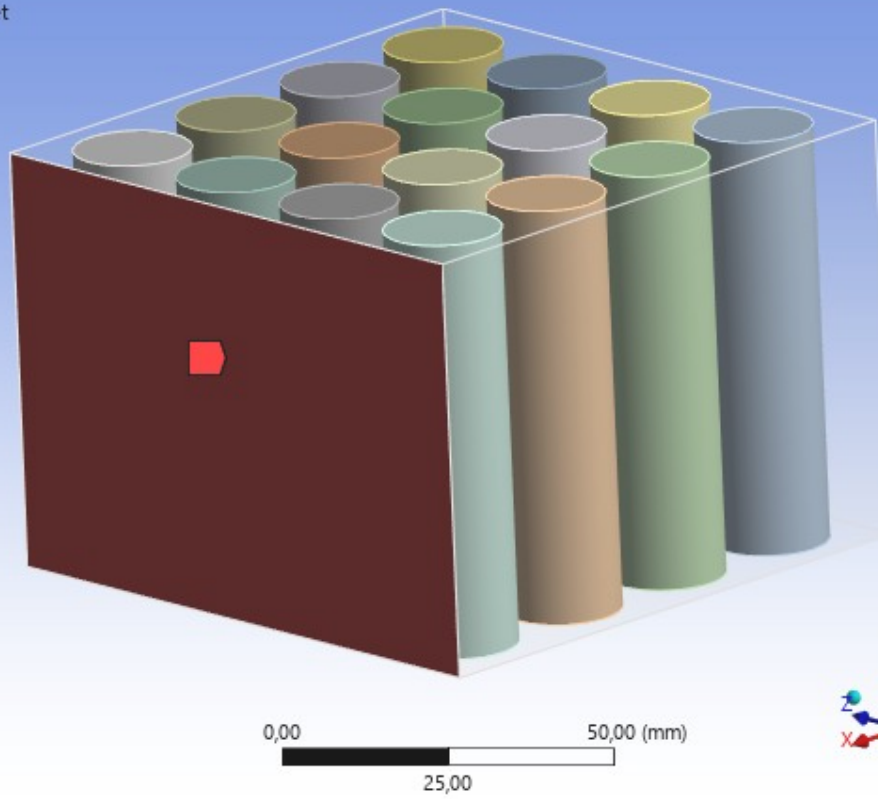


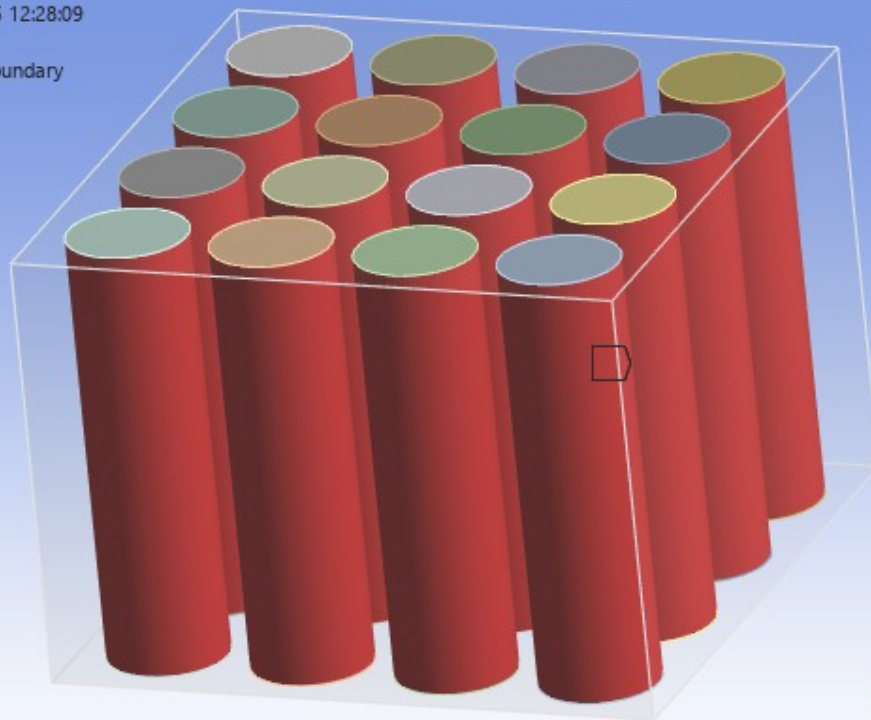
FIGURE 6

Model (A3) > Named Selections > Cell Boundary > Figure

Figure

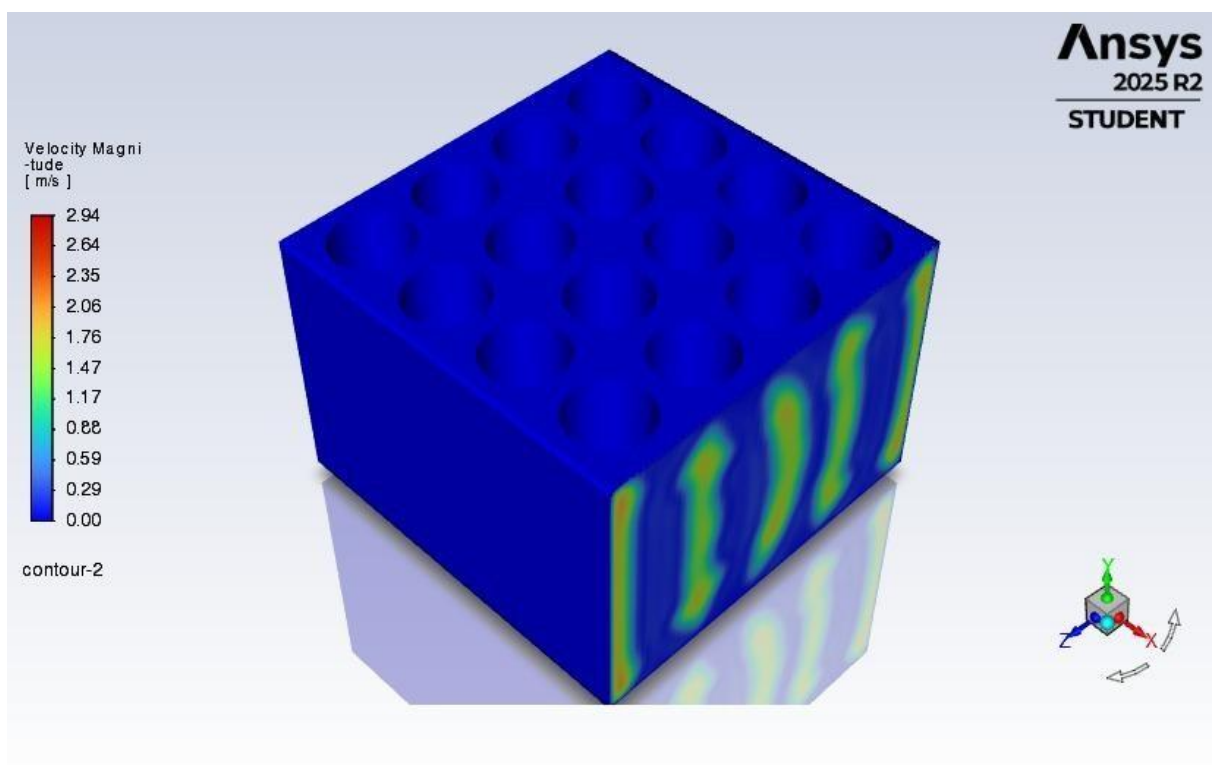
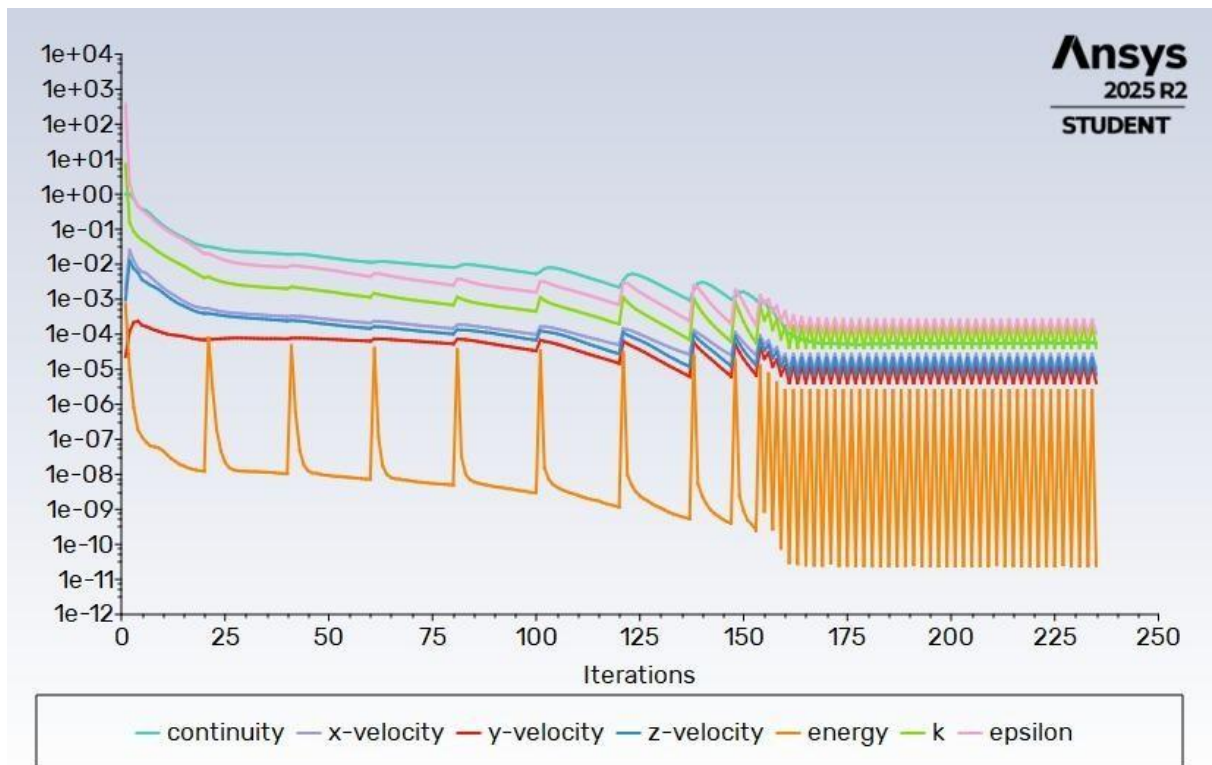
28/10/2025 12:28:09

 Cell Boundary

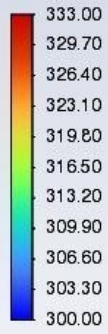


0,00 50,00 (mm)
25,00

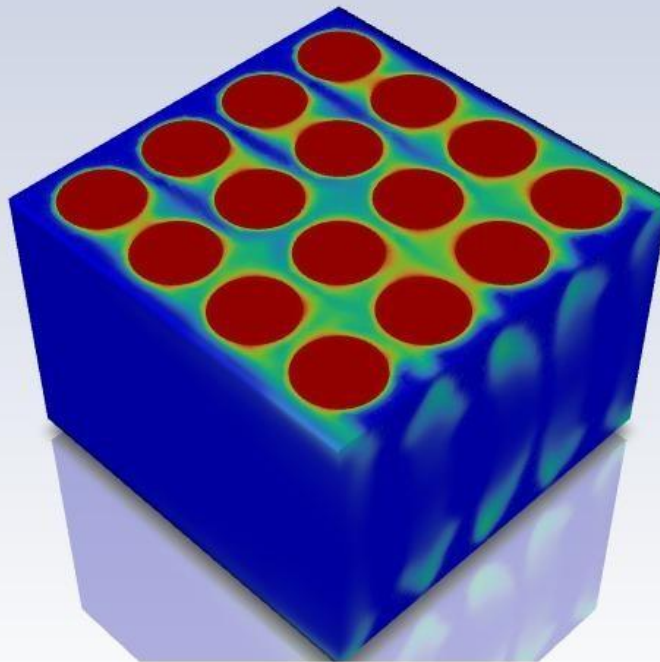
ANSYS FLUID FLOW (FLUENT) IMAGES



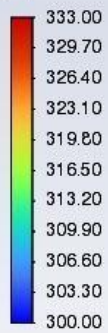
Static Temperature
[K]



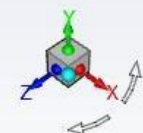
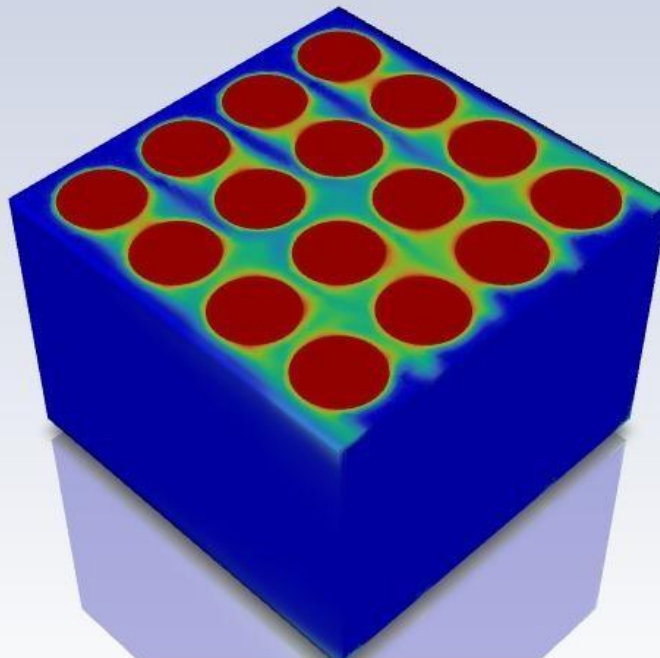
contour-1



Wall Temperature
[K]



contour-3



CFD Post Processing Report

Contents

1. File Report
 - Table 1 File Information for FFF
2. Mesh Report
 - Table 2 Mesh Information for FFF
3. Physics Report
 - Table 3 Domain Physics for FFF
 - Table 4 Boundary Physics for FFF
4. User Data
 - Figure 1
 - Figure 2

1. File Report

Table 1. File Information for FFF

Case	FFF
File Path	D:\Battery Pack Simulation\16 Pack Battery_files\dp0\FFF\Fluent\FFF.flprj
File Date	28 October 2025
File Time	11:04:38 AM
File Type	CFF
File Version	--

2. Mesh Report

Table 2. Mesh Information for FFF

Domain	Nodes	Elements
battery_pack	112770	93324
cell cell_1	16110	15048
cell cell_10	16110	15048
cell cell_11	16110	15048
cell cell_12	16110	15048
cell cell_13	16110	15048
cell cell_14	16110	15048
cell cell_15	15795	14740
cell cell_16	16110	15048
cell cell_2	16110	15048
cell cell_3	16110	15048
cell cell_4	16110	15048
cell cell_5	16110	15048
cell cell_6	16110	15048
cell cell_7	16110	15048
cell cell_8	15795	14740
cell cell_9	16110	15048
All Domains	369900	333476

3. Physics Report

Table 3. Domain Physics for FFF

Domain - battery_pack	
Type	FLUID
Location	battery_pack
Domain - cell cell_1	
Type	FLUID
Location	cell-cell_1
Domain - cell cell_10	
Type	FLUID
Location	cell-cell_10
Domain - cell cell_11	
Type	FLUID
Location	cell-cell_11
Domain - cell cell_12	
Type	FLUID
Location	cell-cell_12
Domain - cell cell_13	
Type	FLUID
Location	cell-cell_13
Domain - cell cell_14	
Type	FLUID
Location	cell-cell_14
Domain - cell cell_15	
Type	FLUID
Location	cell-cell_15
Domain - cell cell_16	
Type	FLUID
Location	cell-cell_16
Domain - cell cell_2	
Type	FLUID
Location	cell-cell_2
Domain - cell cell_3	
Type	FLUID
Location	cell-cell_3
Domain - cell cell_4	

Type	FLUID
Location	cell-cell_4
Domain - cell cell_5	
Type	FLUID
Location	cell-cell_5
Domain - cell cell_6	
Type	FLUID
Location	cell-cell_6
Domain - cell cell_7	
Type	FLUID
Location	cell-cell_7
Domain - cell cell_8	
Type	FLUID
Location	cell-cell_8
Domain - cell cell_9	
Type	FLUID
Location	cell-cell_9

Table 4. Boundary Physics for FFF

Domain	Boundaries	
battery_pack	Boundary - contact_region src	
	Type	INTERFACE
	Location	contact_region-src
	Boundary - contact_region_10 src	
	Type	INTERFACE
	Location	contact_region_10-src
	Boundary - contact_region_11 src	
	Type	INTERFACE
	Location	contact_region_11-src
	Boundary - contact_region_12 src	
	Type	INTERFACE
	Location	contact_region_12-src
	Boundary - contact_region_13 src	
	Type	INTERFACE
	Location	contact_region_13-src
	Boundary - contact_region_14 src	
	Type	INTERFACE

Location	contact_region_14-src
Boundary - contact_region_15 src	
Type	INTERFACE
Location	contact_region_15-src
Boundary - contact_region_16 src	
Type	INTERFACE
Location	contact_region_16-src
Boundary - contact_region_2 src	
Type	INTERFACE
Location	contact_region_2-src
Boundary - contact_region_3 src	
Type	INTERFACE
Location	contact_region_3-src
Boundary - contact_region_4 src	
Type	INTERFACE
Location	contact_region_4-src
Boundary - contact_region_5 src	
Type	INTERFACE
Location	contact_region_5-src
Boundary - contact_region_6 src	
Type	INTERFACE
Location	contact_region_6-src
Boundary - contact_region_7 src	
Type	INTERFACE
Location	contact_region_7-src
Boundary - contact_region_8 src	
Type	INTERFACE
Location	contact_region_8-src
Boundary - contact_region_9 src	
Type	INTERFACE
Location	contact_region_9-src
Boundary - inlet	
Type	VELOCITY-INLET
Location	inlet
Boundary - outlet	
Type	PRESSURE-OUTLET
Location	outlet

	Boundary - wall battery_pack	
	Type	WALL
	Location	wall-battery_pack
cell cell_1	Boundary - cell_boundary contact_region trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region-trg
	Boundary - wall cell cell_1	
	Type	WALL
	Location	wall-cell-cell_1
cell cell_10	Boundary - cell_boundary contact_region_10 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_10-trg
	Boundary - wall cell cell_10	
	Type	WALL
	Location	wall-cell-cell_10
cell cell_11	Boundary - cell_boundary contact_region_11 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_11-trg
	Boundary - wall cell cell_11	
	Type	WALL
	Location	wall-cell-cell_11
cell cell_12	Boundary - cell_boundary contact_region_12 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_12-trg
	Boundary - wall cell cell_12	
	Type	WALL
	Location	wall-cell-cell_12
cell cell_13	Boundary - cell_boundary contact_region_13 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_13-trg
	Boundary - wall cell cell_13	
	Type	WALL
	Location	wall-cell-cell_13
cell cell_14	Boundary - cell_boundary contact_region_14 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_14-trg
	Boundary - wall cell cell_14	

	Type	WALL
	Location	wall-cell-cell_14
cell cell_15	Boundary - cell_boundary contact_region_15 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_15-trg
	Boundary - wall cell cell_15	
	Type	WALL
	Location	wall-cell-cell_15
cell cell_16	Boundary - cell_boundary contact_region_16 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_16-trg
	Boundary - wall cell cell_16	
	Type	WALL
	Location	wall-cell-cell_16
cell cell_2	Boundary - cell_boundary contact_region_2 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_2-trg
	Boundary - wall cell cell_2	
	Type	WALL
	Location	wall-cell-cell_2
cell cell_3	Boundary - cell_boundary contact_region_3 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_3-trg
	Boundary - wall cell cell_3	
	Type	WALL
	Location	wall-cell-cell_3
cell cell_4	Boundary - cell_boundary contact_region_4 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_4-trg
	Boundary - wall cell cell_4	
	Type	WALL
	Location	wall-cell-cell_4
cell cell_5	Boundary - cell_boundary contact_region_5 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_5-trg
	Boundary - wall cell cell_5	
	Type	WALL

	Location	wall-cell-cell_5
cell cell_6	Boundary - cell_boundary contact_region_6 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_6-trg
	Boundary - wall cell cell_6	
	Type	WALL
	Location	wall-cell-cell_6
cell cell_7	Boundary - cell_boundary contact_region_7 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_7-trg
	Boundary - wall cell cell_7	
	Type	WALL
	Location	wall-cell-cell_7
cell cell_8	Boundary - cell_boundary contact_region_8 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_8-trg
	Boundary - wall cell cell_8	
	Type	WALL
	Location	wall-cell-cell_8
cell cell_9	Boundary - cell_boundary contact_region_9 trg	
	Type	INTERFACE
	Location	cell_boundary-contact_region_9-trg
	Boundary - wall cell cell_9	
	Type	WALL
	Location	wall-cell-cell_9

4. User Data

Figure 1.

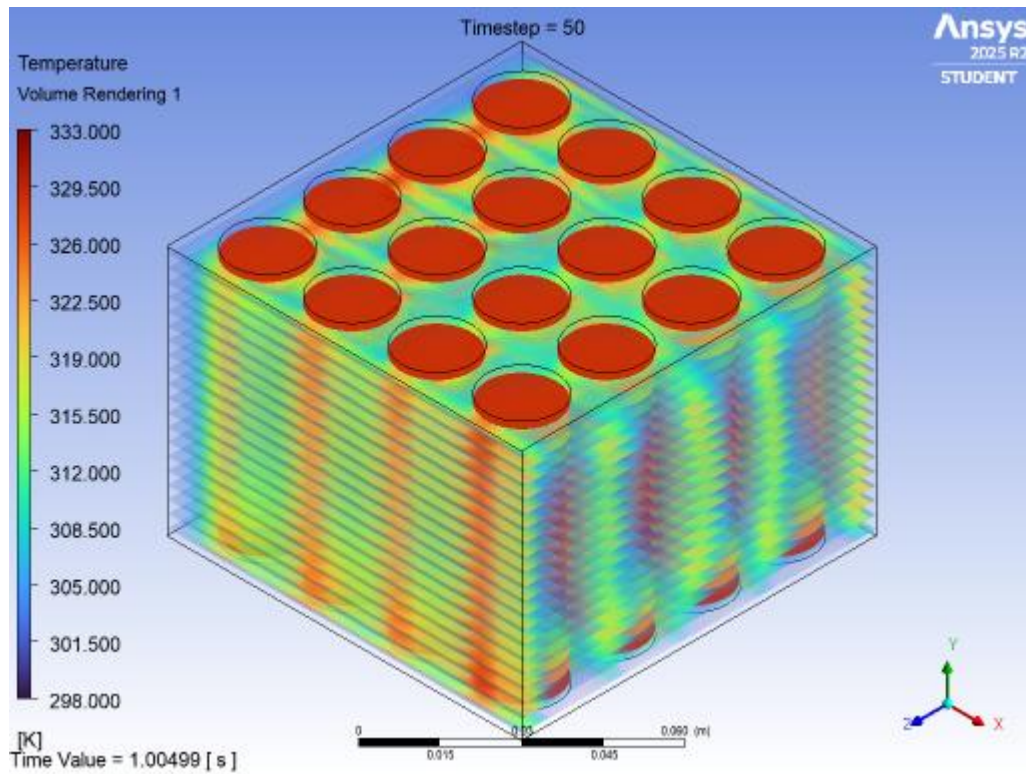


Figure 2.

